

E-Commerce Sales Analysis Project Report

Project Overview and Objectives

This project analyzes an E-Commerce Sales dataset using Python, Pandas, and Matplotlib. It performs data cleaning, sales analysis, and visualizations to generate business insights.

Objectives:

- Analyze sales data
- Clean dataset
- Product-wise and region-wise analysis
- Generate charts
- Produce meaningful insights

Setup and Installation Instructions

Software Used:

- Python 3.x
- Visual Studio Code
- Pandas
- Matplotlib

Install:

```
pip install pandas matplotlib
```

Run:

```
python main.py
```

Code Structure Explanation

Ecommerce-Sales-Analysis/

|

├── main.py

├── sales_data.csv

├── README.md

├── report.md

├── requirements.txt

└── visualizations/

 ├── product_sales.png

 ├── region_sales.png

 └── daily_sales.png

main.py: Loads dataset, cleans data, performs analysis, creates charts.

sales_data.csv: Dataset.

visualizations/: Stores generated charts.

README.md: Project documentation.
report.md: Detailed report.

Screenshots of Working Application

Insert the following screenshots:

1. VS Code showing successful execution.
2. Terminal output.
3. Product Wise Sales Bar Chart.
4. Region Wise Sales Pie Chart.
5. Daily Sales Trend Line Chart.

Explanation of How the Technical Requirements Were Met

Requirement	Status
Dataset Loaded Successfully	✓ Completed
Data Cleaning	✓ Completed
Summary Statistics	✓ Completed
Product-wise Analysis	✓ Completed
Region-wise Analysis	✓ Completed
Daily Sales Analysis	✓ Completed
Bar Chart	✓ Completed
Pie Chart	✓ Completed
Line Chart	✓ Completed
Business Insights	✓ Completed
Professional Documentation	✓ Completed

Project Insights

- Identified the best-selling products.
- Compared regional sales performance.
- Analyzed daily sales trends.
- Generated visual reports for better understanding.

Conclusion

This project demonstrates practical data analysis using Python, Pandas, and Matplotlib. It provides meaningful insights through data cleaning, analysis, and visualization.

```
File Edit Selection View Go Run Terminal Help python
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\komal\OneDrive\Desktop\python> & C:\Python314\python.exe c:/Users/komal/OneDrive/Desktop/python/data.py
Dataset Loaded Successfully!

First 5 Rows:
   Date      Product  Quantity  Price  Customer_ID  Region  Total_Sales
0  2024-01-01    Phone         7   37300    CUST001    East      261100
1  2024-01-02  Headphones         4   15406    CUST002   North      61624
2  2024-01-03    Phone         2   21746    CUST003   West       43492
3  2024-01-04  Headphones         1   30895    CUST004    East       30895
4  2024-01-05    Laptop         8   39835    CUST005   North     318680

Dataset Shape:
(100, 7)

Columns:
Index(['Date', 'Product', 'Quantity', 'Price', 'Customer_ID', 'Region',
       'Total_Sales'],
      dtype='str')

Missing Values:
Date          0
Product       0
Quantity      0
Price         0
Customer_ID   0
Region        0
Total_Sales   0
dtype: int64

Dataset after Cleaning:
(100, 7)

Summary Statistics:
      Quantity      Price      Total_Sales
count  100.000000  100.000000  100.000000
mean     4.780000  25808.510000 123650.480000
std     2.588163  13917.630242 100161.085275
min     1.000000  1308.000000  6540.000000
25%     2.750000  14965.250000  39517.500000
50%     5.000000  24192.000000  97955.500000
75%     7.000000  38682.250000 175792.500000
```

```
File Edit Selection View Go Run Terminal Help python

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\komal\OneDrive\Desktop\python> & C:\Python314\python.exe c:/Users/komal/OneDrive/Desktop/python/data.py
Dataset after Cleaning:
(100, 7)

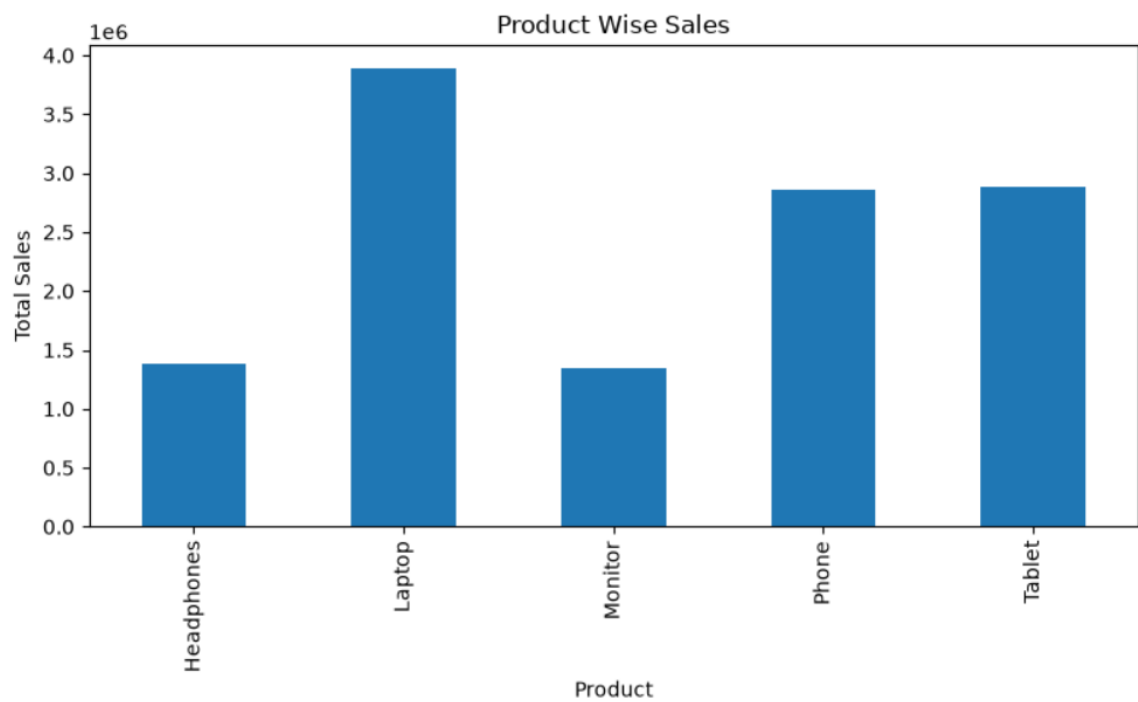
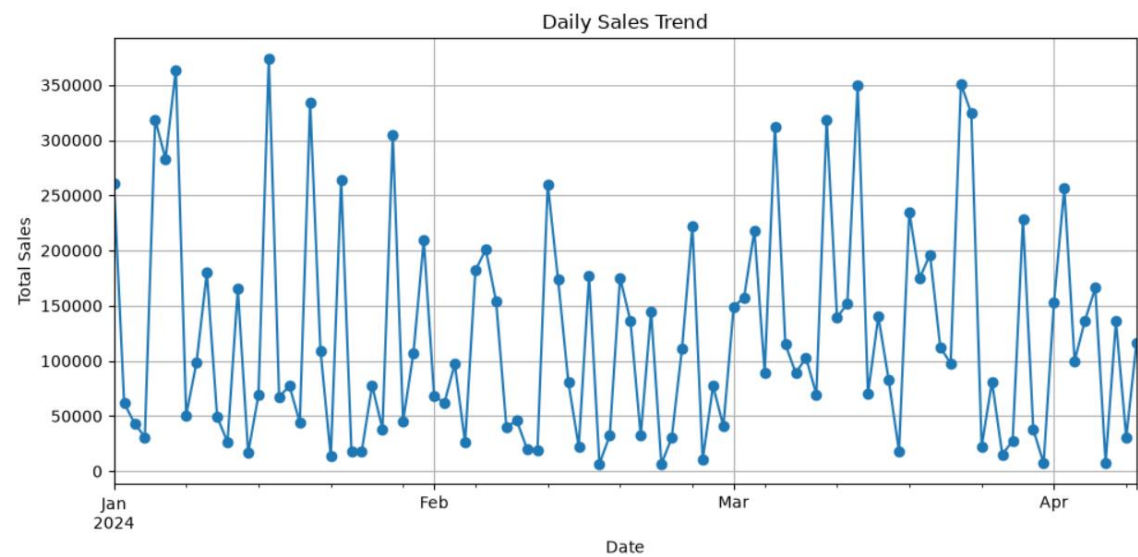
Summary Statistics:
      Quantity      Price      Total_Sales
count  100.000000   100.000000   100.000000
mean    4.780000  25808.510000  123650.480000
std     2.588163  13917.630242  100161.085275
min     1.000000   1308.000000   6540.000000
25%     2.750000  14965.250000  39517.500000
50%     5.000000  24192.000000  97955.500000
75%     7.000000  38682.250000  175792.500000
max     9.000000  49930.000000  373932.000000

Total Sales = 12365048
Average Sales = 123650.48
Highest Sale = 373932
Lowest Sale = 6540

Product Wise Sales
Product
Headphones    1384033
Laptop        3889210
Monitor       1348071
Phone         2859394
Tablet        2884340
Name: Total_Sales, dtype: int64

Region Wise Sales
Region
East    2519639
North   3983635
South   3737852
West    2123922
Name: Total_Sales, dtype: int64
```

Visualization



Region Wise Sales

